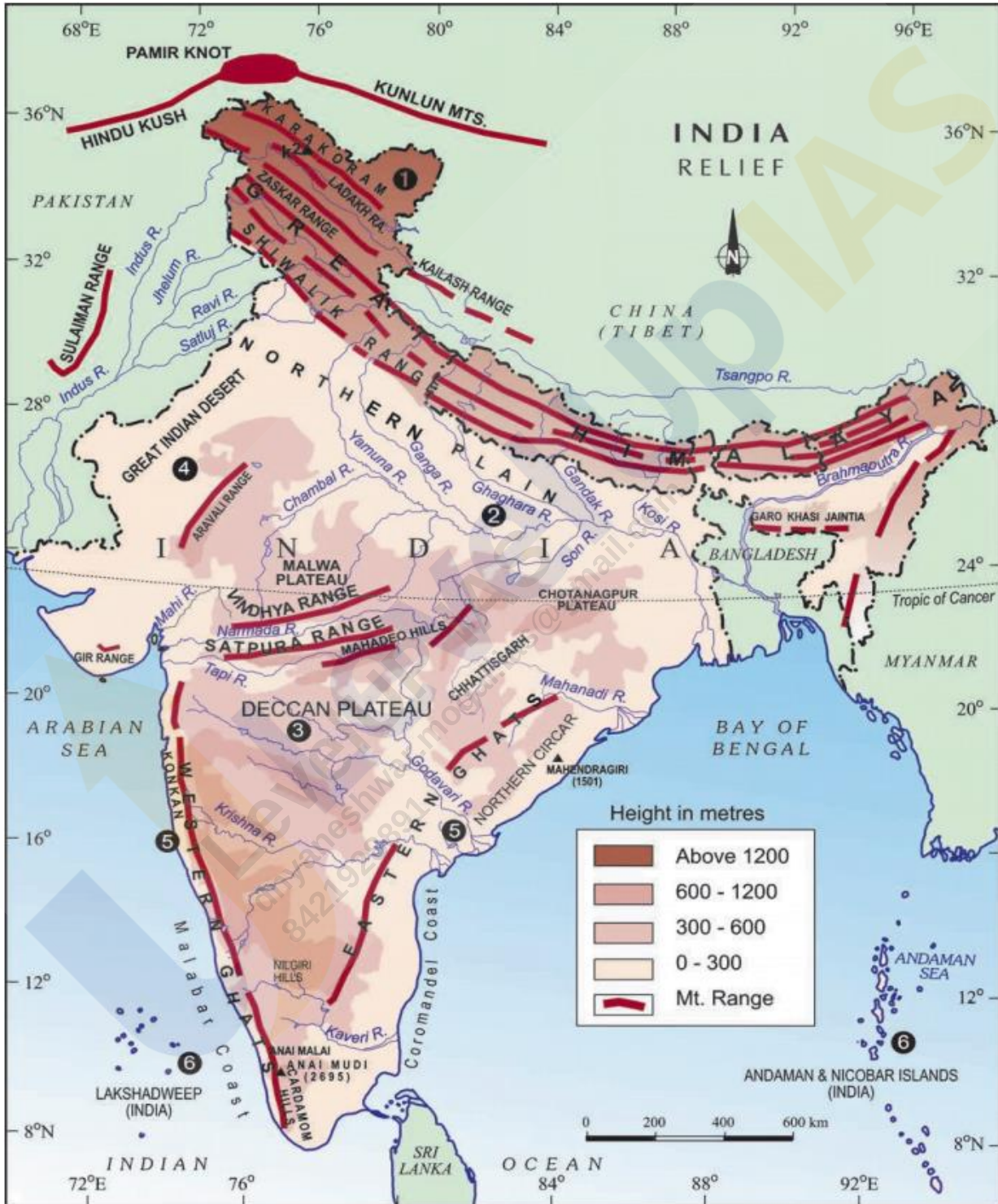


INDIAN PHYSIOGRAPHY

Physiography deals with physical features of the earth's surface.

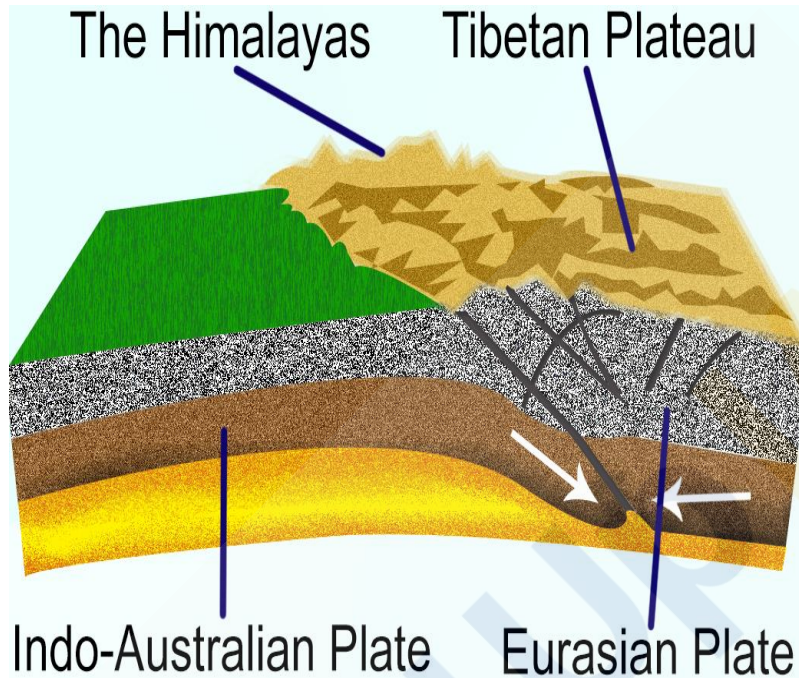
The physiography of India is divided into 05 major feature:

- (i) The Himalayan mountains
- (ii) The Northern plains
- (iii) The Peninsular plateau
- (iv) The Indian Desert
- (v) The Coastal plains
- (vi) The Islands



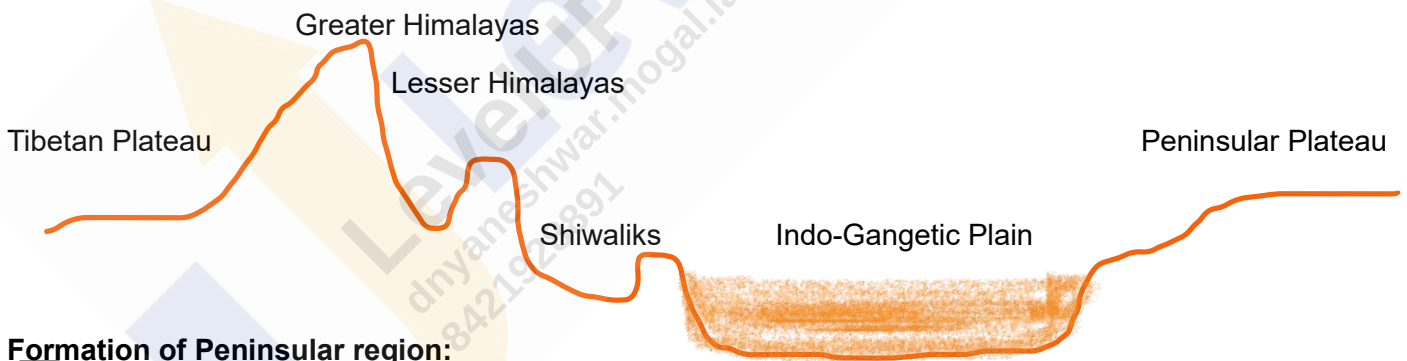
Formation of Himalayas:

- The Gondwana land consists of – India, Australia, S.Africa, S.America & Antarctica
- Indo-Australian plate started drifting towards north because of conventional currents.
- This drifting led to collision with much larger plate i.e., Eurasian plate.
- This collision led to accumulation of sedimentary rocks in geosyncline known as Tethys, this resulted in folding, to form the mountain system of western Asia and Himalaya.



Formation of Northern Plains:

- Upliftment of Himalaya out of Tethys sea + subsidence of Northern flank = large basin formed
- This depression filled with deposition of sediments by rivers.
- A flat based region of widespread alluvial accumulations prompted the creation of Northern plains.

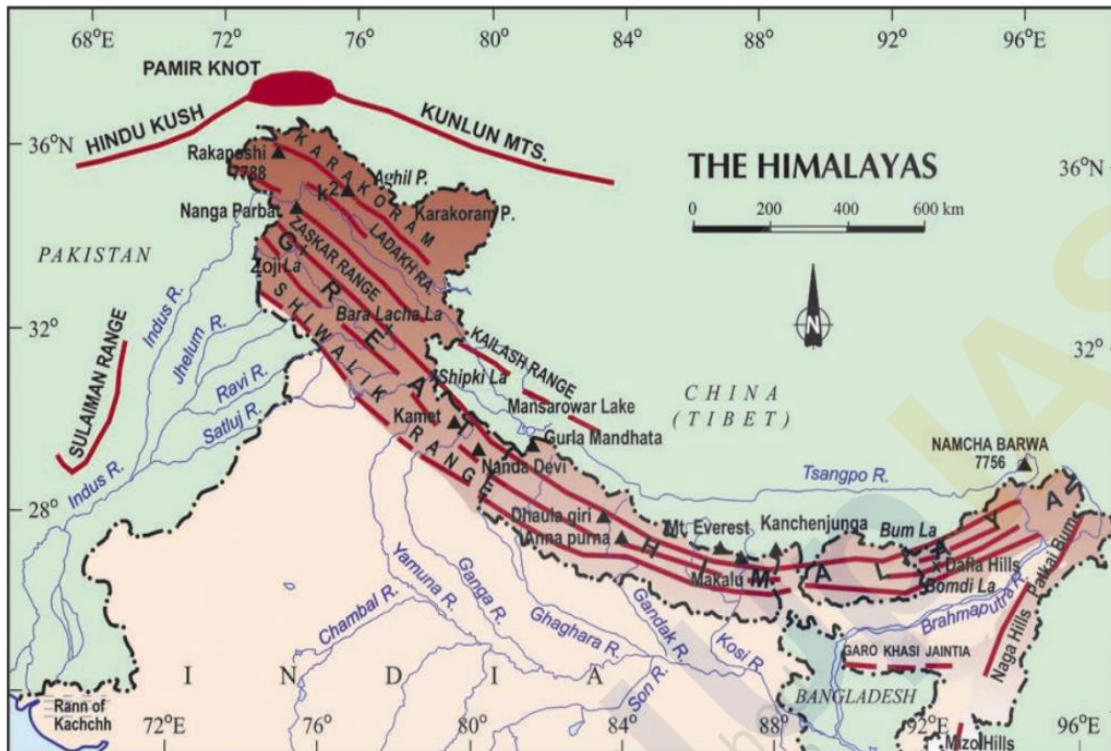


Formation of Peninsular region:

- Formed due to breaking & drifting of Gondwana land.
- Tectonically stable
- Mix of Gondwana, Archean, cuddapah and Dharwar rocks
- Made of series of patland plateaus (Hazaribagh, Palamu, Ranchi, Malwa, Coimbatore, Karnataka)
- Composed of igneous and metamorphic rocks
- Gentle rising hills & wide valleys.

The Himalayas

- ➔ From N-W to S-E.
- ➔ Approx. length (central axial range) is 2500 km (E-W)
- ➔ Divided into 03 mains mountain ranges:
 - (i) Greater Himalayas/Himadri
 - (ii) Lesser Himalayas/ Himachal
 - (iii) Outer Himalayas / Shiwaliks



1. **Greater Himalayas:**
 - ➔ Most continuous range
 - ➔ Consists of loftiest peaks & all prominent Himalayan peaks.
 - ➔ Steeper slope towards south & gentle slope towards north.
 - ➔ Snowline height is more in west than in east.

Important peaks in Himalayas:

Mount Everest – highest, also known as Sagarmatha

Kanchenjunga	Makalu	Dhaulagiri	Nanga parbhat	Annapurna
Nanda devi	Badrinath	kedarnath		

2. **Lesser Himalayas:**
 - ➔ Range ie. South of Himadri
 - ➔ Most rugged mountain range
 - ➔ Composed of highly compressed and altered rocks.
 - ➔ Altitude – 3700-4500 mts

Important ranges:

Pir Panjal – Longest & most important Dhauladhar Mahabharat ranges

Important valleys:

Kashmir valley Kangra valley Kullu valley

3. **Outer Himalayas:**
- Composed of unconsolidated sediments by rivers
 - Valleys are covered with thick gravel & alluvium
 - Extends from potwar basin to kosi river (W-E)

Duns: longitudinal valley lying between lesser Himalaya and the Shiwaliks

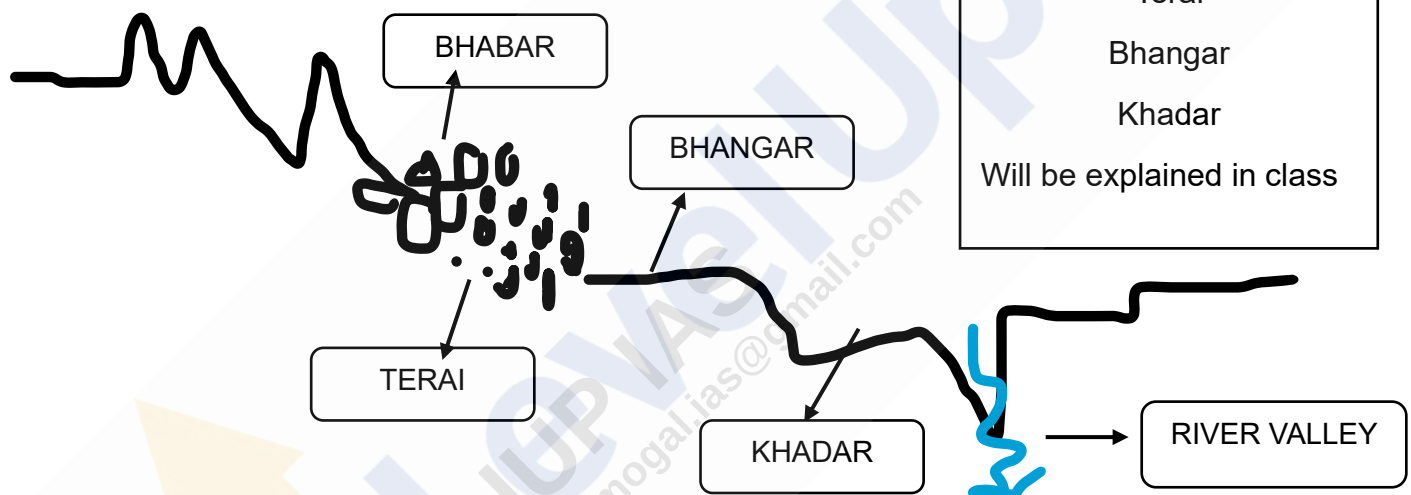
Dehra dun Kotli dun Patli dun

Duars: Terai areas in West Bengal (only nitrate rich plain in India)

Western Duar – Bengal Duar Eastern Duar- Assam Duar

Northern Plains

- Formed due to 03 rivers (The Indus, The Ganga, The Bhramaputra)
- Alluvium lying at the foothills of Himalayas formed this fertile land.
- Densely populated
- More agricultural productivity



Other Plains of India:



Peninsular India

- A Triangular region characterised with east ward gentle slope because of east ward movement of the peninsular rivers.
- Oldest landform.
- Large mineral depositions
- Constitutes several smaller plateaus & hills.

Various hills of peninsula region:

Aravalli mountain range

Vindhyaachal mountain range

Western ghats

Eastern Ghats

Satpura range

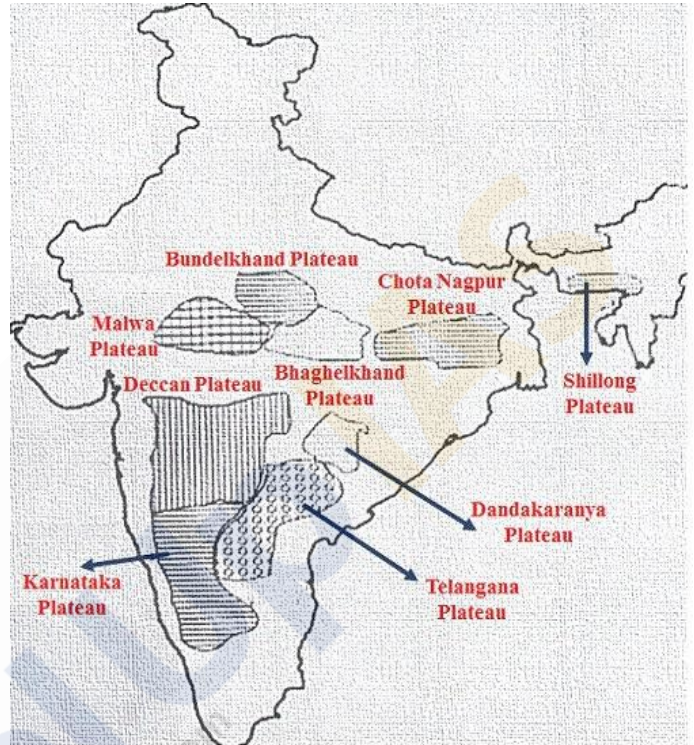
Passes in western Ghats (north – south):

Thal Ghat

Bhor Ghat

Pal Ghat

Senkotta pass



Indian Desert

- Desert is an arid & semi-arid region.
- The great Indian desert – The Thar Desert (N-W of Aravali hills)
- Semi-arid region of the Thar desert – Bagar
- Covered by sand plains consists of sand dunes.
- Sand dunes & mushroom rocks are geomorphological features.

Formation of Indian Desert:

- Initially it's a rock structure of peninsular plateau.
- Due to extreme climatic condition, it turned into desert.

Conditions for deserts formation:

Off shore winds
Rain shadow region
Sub-tropical regions
Continentality
Cold ocean currents



Coastal plains

- Divided as western & Eastern coastal plain.
- Fertile soils.
- Known for ports
- Mineral oil deposits
- Monazite deposits
- Fishing and salt production

- (i) Eastern Coastal Plain:
- Between eastern ghats & Bay of Bengal
 - Godavari, Krishna, Mahanadi Kaveri deltas are found.
 - Chilika & Pulicat lake are important.
 - Called as Utkal coast, Coromandel coast, Northern Circars.
 - Emergent coast – upliftment of land due to depositional activity.
 - Extensive plains.
- (ii) Western Coastal Plain:
- From Gulf of Camba to Cape Camron
 - Divided into Konkan coast, Karnataka coast, Kerala coast.
 - Known for coves (very small bays)
 - Vembanad lake – largest lake
 - Submergent coast – faulting
 - Narrow

Indian Islands

Major island of India are Andaman & Nicobar in BOB, Lakshadweep in sea.

- (i) Andaman & Nicobar :
- Formed due to collision of Indian plate & Burma minor plate.
 - Southward extension of Arakanian ranges.
 - Duncan Passage separates Little Andaman & South Andaman.
 - 10° channel separates Andaman & Nicobar.
- (ii) Lakshadweep:
- Formed due to deposition of corals.
 - 8° channel separates islands of Minicoy & Maldives.
 - These are 03 types viz., Atoll islands, Laccadive islands, Minicoy islands. (Minicoy – largest)

